

TrES-3b

R-1353

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_160413 · created 2026-07-31T22:50:57+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457491.8667 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.138 +/- 0.013
Transit depth (Rp/Rs)^2	1.91 +/- 0.36 [%]
Orbital Inclination (inc)	83.79 +/- 0.74
Airmass coefficient 1 (a1)	1.0309 +/- 0.0082
Airmass coefficient 2 (a2)	-0.0216 +/- 0.0023
Scatter in the residuals of the lightcurve fit is	0.79 %
Best Comparison Star	#2 - [81, 268]
Optimal Aperture	3.51
Optimal Annulus	7.71
Transit Duration (day)	0.0666 +/- 0.0046

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.138 ± 0.013 vs 0.16309 (deviation 1.52σ)
Duration fit vs published	0.0666 d vs 0.0567 d (deviation 1.7σ)
Mid-time O–C	-3.5 min
Depth SNR	8.4
Residual scatter	0.79 %

Light curve

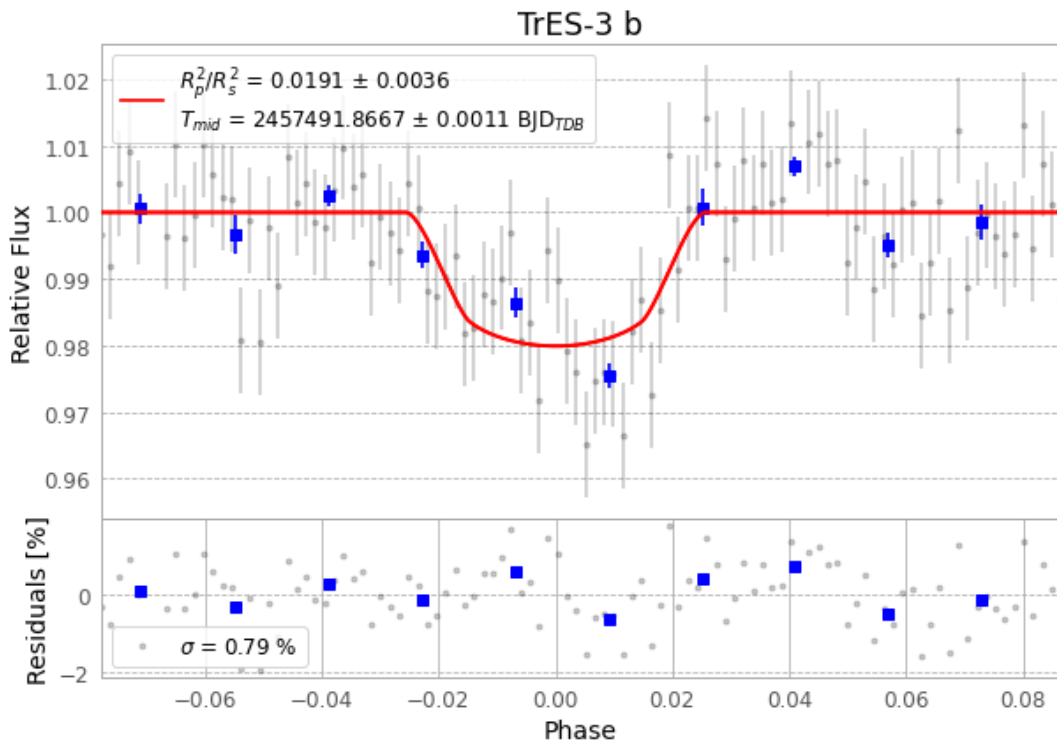


Figure 1 — FinalLightCurve_TrES-3 b_2016-04-12.png (SHA-256 9c34df3aa2e13db6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [280, 175], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 374f02e5cfec8655...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

104 FITS frames (68 MB), TRES-3160413061812.FITS → TRES-3160413112712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-160413.log (c6d6337d7968...), FinalLightCurve_TrES-3 b_2016-04-12.png (9c34df3aa2e1...), FinalLightCurve_TrES-3 b_2016-04-12.csv (15a3c129f930...)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 22:50Z.